



The monkey on the Rope

The Puzzle

Spoilers ahead! - This page contains the answers to the puzzle. If you haven't already come up with your own answers, you're in the wrong place. Please see the description of the [Monkey Puzzle](#) first, think about it, and *then* read this page!

The Initial Situation

First of all, we'll look at the initial equilibrium situation, before the monkey starts to try to climb. The monkey's weight acts downwards, we'll call it w . This pulls the rope with a force w , and the rope transfers this force directly to the weight on the other end of the rope, pulling the weight upwards with a force w . However, the weight also pulls downwards with an equal force w , which pulls the rope with a force w , which acts on the monkey. Therefore both the weight and the monkey are pulled downwards by their own weights, and pulled upwards by the rope with equal forces. This is why it's in equilibrium, and is why neither the monkey or the weight move. Another way of saying all this is to say that the monkey balances the weight, so they don't move.

The easy bit - the weight

Now, what happens when the monkey tries to climb the rope? The monkey exerts an additional force on the rope, so it pulls the rope down with a force which is now greater than w . How is this possible? In exactly the same way as any person would climb any rope - when they hang from a rope, the rope is pulled by their weight, but when they climb it, they pull the rope with a larger force. Hopefully, whatever is supporting the rope is strong enough not to break under this extra force and they can climb the rope. Tree branches bend much more when being climbed on than when being hung from. [Play safe, kids]. This exertion of a greater force than one's weight happens every time someone does a chin-up, or press-up, or in fact just stands up from the sofa.

OK, maybe we've laboured that point. The monkey pulls the rope. On the other end of the rope, the weight is now pulled upwards not just by the monkey's weight w , but by its climbing force too. It is only pulled downwards by its own weight, w , though, so the net force is upwards. Therefore the weight will accelerate upwards.

All those who said the weight goes downwards, or stays where it is, go back to the [puzzle](#) and rethink your answer. Now, what happens to the monkey?

The harder bit - the monkey

So now we think we know what happens to the weight, but what about the monkey? We still have our three possibilities, as follows:

- The monkey goes upwards
- The monkey stays still
- The monkey goes downwards

The monkey goes up

Argument for: Action and Reaction. When the monkey exerts its climbing force, it experiences an equal reactionary force upwards, and so has the same force upwards as the weight does. Therefore, since they have the same mass, the monkey climbs upwards with the same acceleration as the weight.

Argument against: Action and Reaction doesn't apply. If the rope was fixed, then the support would be able to provide this additional force to lift the monkey up, but if the other end is not fixed, and it's moving, then there's nowhere for this extra force to come from. Let's compare it with the following diagram:

Here we have the same monkey on a fixed rope. Again, in the equilibrium position, gravity pulls the monkey down with the same force w , and the support of the rope provides a reactionary force w . Again, everything is static. However, once the monkey tries to climb, it exerts its additional climbing force, the rope exerts this force on the support, and because the support is fixed, it provides an equal and opposite reaction. Therefore when the monkey exerts its climbing force F , it accelerates upwards with an acceleration F/m (where m is the mass of the monkey).

That much is easy. Now, let's go back to the original problem. When the rope is *not* fixed to a rigid support, and it's moving over the pulley, there's no way that the monkey can climb at the *same rate* as on a fixed rope. If both monkey and weight are now accelerating upwards with this same acceleration F/m , that *must* be harder than just climbing a fixed rope, that clearly *must* require more force. In energy terms, now we have both the weight and the monkey rising ever faster upwards, requiring twice as much potential energy, and twice as much kinetic energy as the fixed rope case, but yet this argument says that the monkey is still pulling with the same force for both cases. Double the energy? That just doesn't make sense.

The monkey stays still

Argument for: All forces balance. As discussed above, the weight is incapable of providing a reactionary force (as the firmly-fixed rope would be able to do), so the weight still pulls on the rope with the same force w . This balances the monkey's weight, and so the monkey remains stationary (passing the rope downwards). When we compare this with the fixed rope case, we see that the only difference is that the weight rises instead of the monkey - the monkey exerts the same effort but it is just diverted into raising the weight instead.

Argument against: Angular momentum. In our system, angular momentum (about the pulley) must be conserved, and at the beginning, when everything is stationary, obviously the total angular momentum is zero. If the weight is rising (and accelerating), that possesses angular momentum about the pulley (its velocity multiplied by the radius of the pulley), and if the monkey remains stationary, then angular momentum is not conserved. So this argument can't be right.

The monkey goes downwards

Argument for: The weight is accelerating, so no matter how hard the monkey tries to climb, he can't keep up with it, and goes downwards with respect to the pulley.

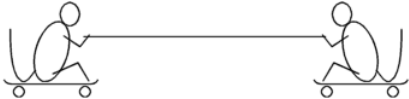
Argument against: For the monkey to be forced downwards, there must be a net downwards force on it. Its weight is constant, and there's no way for the force from the rope to be reduced (it's still got the weight on the other end), so there's no way for there to be a net downward force on the monkey. So therefore the monkey can't go downwards (unless he tries to climb *down* the rope???)

Also, from this argument, our monkey could climb happily up to a certain speed, and then stop climbing. This argument doesn't tell us anything about what happens before everything reaches the monkey's maximum speed. As to what happens when the monkey *does* stop climbing, we'll come to that later.

Action and Reaction

Newton's third law is often quoted, but not quite so often clearly understood. To make things a little simpler, let's take away the gravity part of our problem, and arrange it horizontally. We'll even give our monkey a friend, and a pair of skateboards:

Initially, there's no force on the rope and both monkeys are stationary. Now let one of the monkeys pull on the rope. Obviously the other monkey will be pulled towards the centre, but what about the pulling monkey? Here in this simplified picture it's a lot clearer that he also must be pulled towards the centre, and in fact it doesn't matter which one pulls. The equal and opposite reaction doesn't "come from" the other monkey, it is caused simply by the pulling. Whether the rope is fixed or not, moving or not, the action and the reaction are inseparable. Also note that linear momentum is conserved here, as it is always zero no matter how fast the monkeys hurtle towards each other [Play safe, monkeys].



Two monkeys on skateboards, with a rope

So, we come to the conclusion that the reaction force does not require a fixed rope, and in our original problem, our monkey will indeed experience a reactionary force as a result of its attempts to climb.

Angular Momentum

Angular momentum at first seems an inappropriate concept here, because we haven't got anything spinning (except perhaps a massless pulley). But we can usefully examine the angular momenta of the monkey and the weight about the pulley, as long as the total angular momentum of the system is conserved. But is it?

This diagram shows an example of where angular momentum is *not* conserved. One weight is attached to one end of the rope, and two identical weights are attached to the other. Initially, the weights are held steady, and the total angular momentum is zero. When the system is let loose, because it is not balanced, the pair of weights accelerate downwards and the single weight is pulled upwards. The total angular momentum is then the sum of all the individual angular momenta (mass times velocity times pulley radius), as they all act in the same direction (here anticlockwise). So after starting at zero, the angular momentum of the system increases and increases - and clearly is not conserved. So why should it be conserved in our original problem?

In fact, angular momentum is only conserved when there is no net torque on a system. If there is a net torque applied from outside, then that torque is the rate of change of angular momentum in our system. For this unequal weights problem, there *is* a net torque on the system, provided by the gravity of the Earth. This net torque, $(2wr - wr)$ acts to increase the angular momentum of our system, as we have seen, and is countered by an equal and opposite angular momentum gained by the Earth about the pulley. Of course, it would be extremely difficult to detect this effect on the Earth itself, but it would be there.

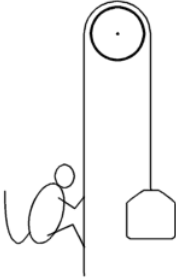
[Incidentally, if the net force on the pair of weights is $(2w-w)$ downwards, and the net force on the single weight is $(2w-w)$ upwards, why doesn't the single weight accelerate twice as fast as the pair of weights, given that it only has half the mass?]

Now back to our original problem. Here, the Earth's gravity acts equally on both sides of the pulley, there is no net torque applied to our system, so the total angular momentum must be conserved.

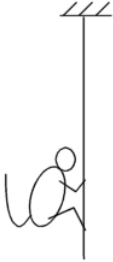
The Final Solution

Our discussion of Action and Reaction seems to point strongly towards answer 1 - the monkey climbs upwards. Our discussion of Angular Momentum also rules out the alternative answers 2 and 3. But is that the end of the story? What about the comparison with the fixed rope?

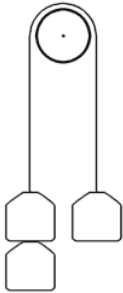
It certainly seems counter-intuitive at first, but we have a somewhat distorted view of effort, force and work when it comes to muscle power. It seems that the monkey is getting energy for free, but that's not the case - it actually has to work twice as hard in order to exert the same force as before. The energy expended is equal to the force multiplied by the distance - in this case the distance is how far the rope has moved relative to the monkey, not how far the monkey has moved relative to the pulley. In order for the monkey to climb a given distance, it must pull twice the distance of rope, so requiring twice as much energy input from the monkey. In that case it's not so unexpected that twice the kinetic and potential energies are thereby gained.



A monkey, a rope, a pulley and a weight. Simple, eh?



A monkey hanging on a fixed rope



Unequal weights connected by rope and pulley

So we've finally arrived at our answer. The weight rises, with a constant acceleration, and the monkey also rises, with an equal acceleration. A similar thing happens when the monkey tries to climb *downwards* too, both weight and monkey mirror each other exactly. The monkey can't get away from the weight.

Is that it?

But we've still got an accelerating weight, an accelerating rope and an accelerating monkey - that still doesn't seem right. Who can climb a rope with their speed constantly accelerating? Something must be wrong. But the constant acceleration just comes from our original assumption, that the monkey is capable of exerting a given climbing force. Given this assumption, everything accelerates constantly, but obviously there will come a point where it either requires too much energy input or too much arm speed to maintain this force. And what happens then?

Let's say that after accelerating up to a certain speed, the monkey gets tired and decides to stop. Now we've got a weight moving upwards at a given speed, and the monkey hanging on the rope. The monkey is no longer trying to climb, so it's just exerting its weight on the rope, and there are no net forces any more. Does that mean that there's nothing to slow the weight down, and that it continues to rise up at its previous speed? Now that the monkey is hanging on the rope, does this mean that it falls with the rope? In that case it *could* get away from the weight, by climbing for a bit and then stopping, but wouldn't that violate the conservation of angular momentum again? Or maybe everything just stops where it was, but in that case what happened to the kinetic energy?

Well let's say that our monkey has a *preferred* climbing speed, rather than a desire for constant acceleration. Is it possible for the monkey to climb at a constant speed? What does this mean in terms of forces? Well, once the weight has reached this speed, the only way to keep it at this constant speed is to remove the climbing force from it. Then it will have no net force on it, and it can continue at a constant speed. But if the monkey stops applying its climbing force, isn't that the same as stopping climbing, which we've just looked at? And if the monkey stops applying its climbing force, then that means it's not exerting any effort any more, so where does the increasing potential energy of the weight come from? And if the weight is rising at a constant speed, then from the discussion earlier, the monkey must also be rising at a constant (and equal) speed too, so how does that work?

The rest is left as an exercise for the reader. Or should that be *"activity"*?

Feedback

Please feel free to submit your thoughts on this problem via email to mail@activityworkshop.net.